DAT158 – Machine learning and advanced algorithms
Institutt for data- og realfag – Høgskulen på Vestlandet

Exercise number 4

Problem 1 (binpacking)

Given part of the instance I = {0.68, 0.67, 0.66, 0.65, 0.63, 0.63, 0.60, 0.60, 0.60, …, | 0.42, 0.42, 0.41, 0.40, 0.40, 0.39, 0.39, 0.39, 0.38, 0.38, 0.38 |… | 12 x 0.2, 0.19}. Show the two groups in beginning, in the middle and at the end obtained by Harmonic grouping (possible that not all the element shown will be present) . The vertical bar indicates the start of a new group. Then show the elements in instance I' coming from these groups.

Problem 2 (Part of compulsory)

a) Implement the randomized algorithm for MAX SAT where we set each x_i to true independently with probability ½ (it is very simple).

b) Assign values to $x_1, \dots, x_5$ by the derandomized version of the algorithm based on methods of conditional expectations (page 103 – 104) to the following clauses

$$C_1 = 3(x_1 \vee \overline{x_2} \vee x_3)$$
$$C_2 = 4(x_1 \vee \overline{x_3})$$
$$C_3 = 1(\overline{x_1} \vee x_2 \vee \overline{x_5})$$
$$C_4 = 3(\overline{x_1} \vee x_3)$$
$$C_5 = 2(x_2 \vee \overline{x_4})$$
$$C_6 = 7(\overline{x_2} \vee x_5 \vee x_9)$$
$$C_7 = 2(x_3 \vee \overline{x_4})$$
$$C_8 = 3(\overline{x_4} \vee \overline{x_5} \vee x_{21})$$
$$C_9 = 5(x_5 \vee \overline{x_{12}})$$
$$\dots$$

The clauses $C_1, \dots, C_9$ are the only clauses containing the variables $x_1, \dots, x_5$.

Problem 3 (Part of compulsory)

We have the following MAX SAT problem

$$C_1 = 4(x_1 \vee \overline{x_3}),\ C_2 = 3(\overline{x_1} \vee x_4), C_3 = 2(\overline{x_2} \vee x_4), C_4 = 1(\overline{x_2} \vee \overline{x_3} \vee x_4)$$

a) Find the ILP (IP) formulation of the problem above.

b) Find the LP formulation of the problem.